

Basis of Animal Classification

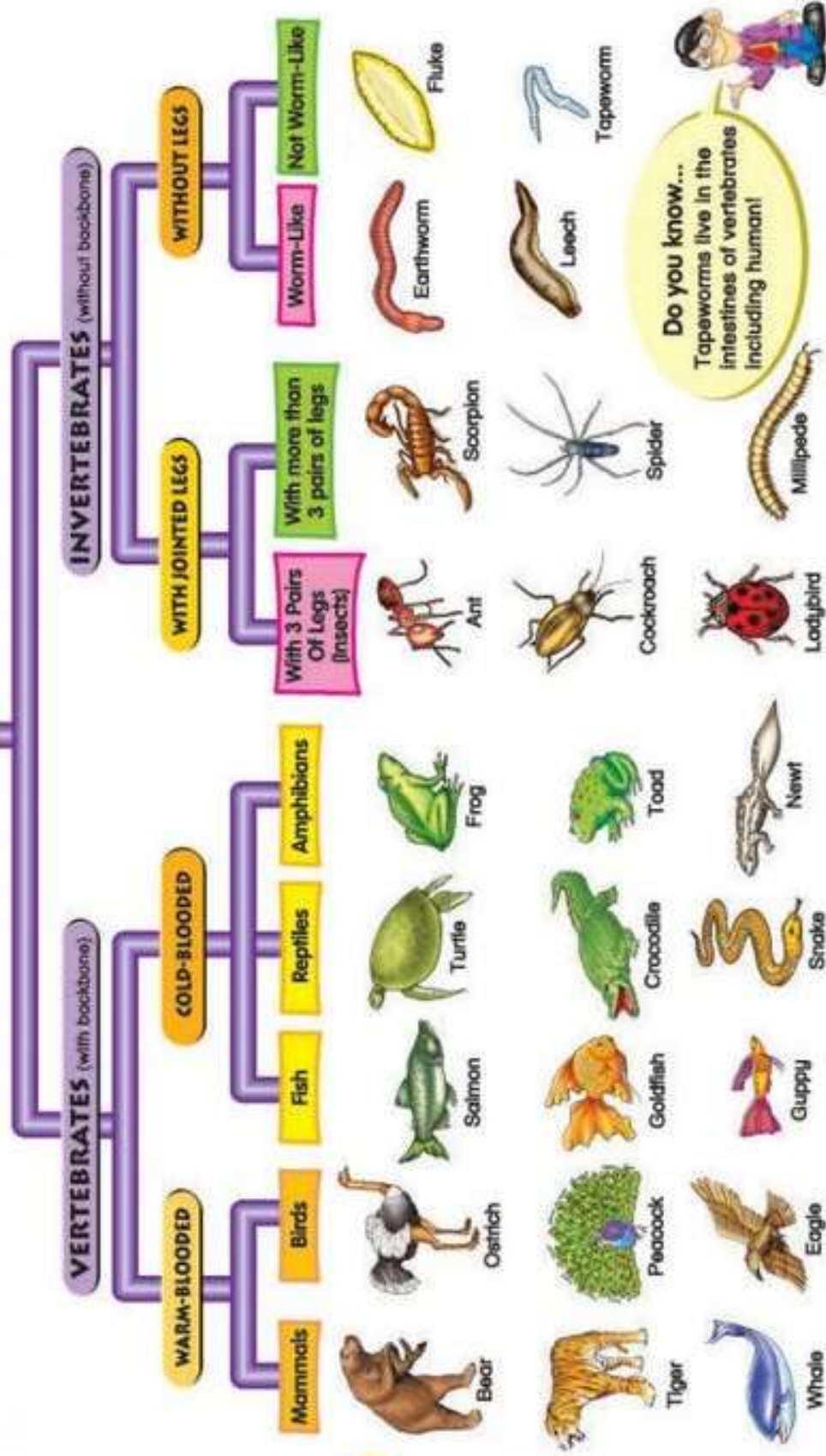
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- The main difference between an embryo and a fetus lies in their stage of development during pregnancy.
- An embryo is the developing baby from fertilization until about the 10th week of pregnancy.
- After that, during the remaining weeks of pregnancy until birth, it is referred to as a fetus.
- The embryonic stage is characterized by the formation of major body structures, while the fetal stage focuses on growth and maturation of those structures.

CLASSIFICATION OF ANIMALS

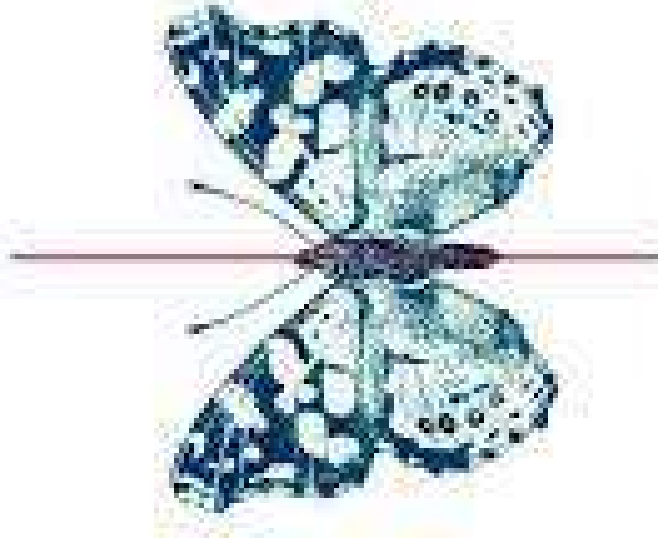


- The animal kingdom is the largest kingdom amongst the five kingdoms consisting of all animals.
- Animals are multicellular eukaryotes and same heterotrophic mode of nutrition.
- Besides these similarities, they are also related to their cell arrangement, body symmetry, the level of organization, coelom, presence/absence of notochord, etc.
- Based on these features, the animal kingdom has been classified into different phyla.

- The different basis for animal classification are:
- Levels of organization
- Symmetry
- Coelom
- Metamerism
- Germ Layers

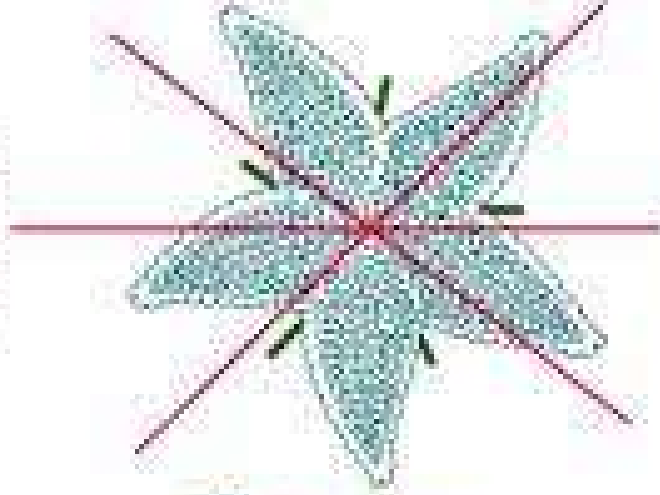
Symmetry

Bilateral Symmetry



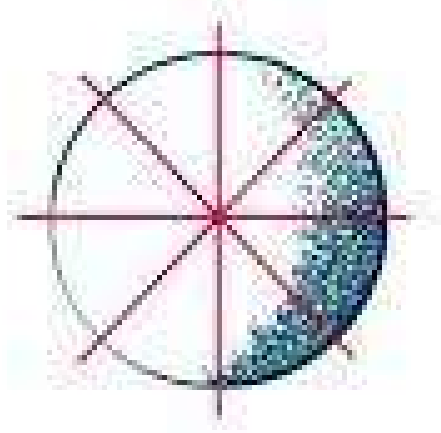
Butterfly (insect)

Radial Symmetry



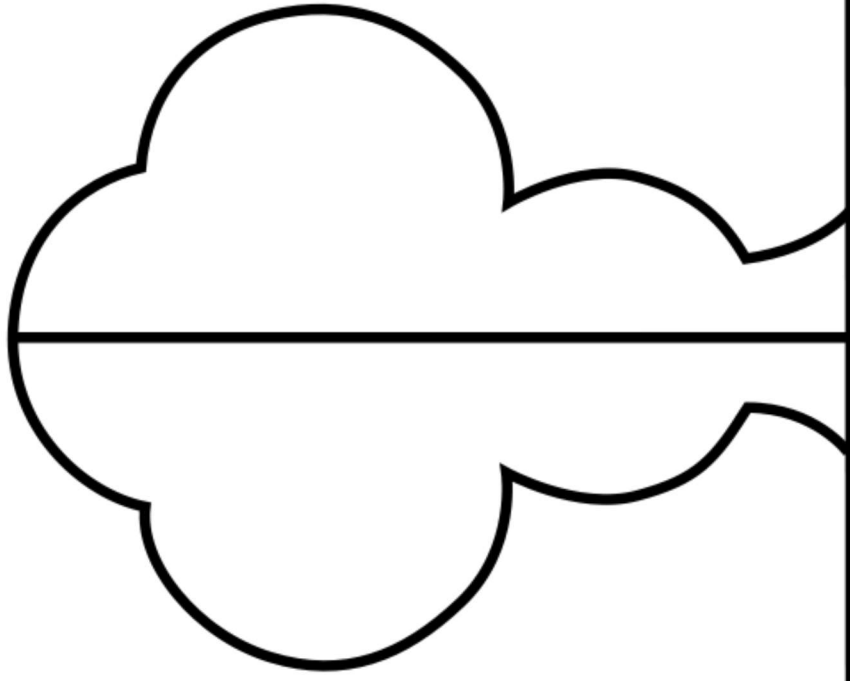
Hydra (invertebrate)

Spherical Symmetry

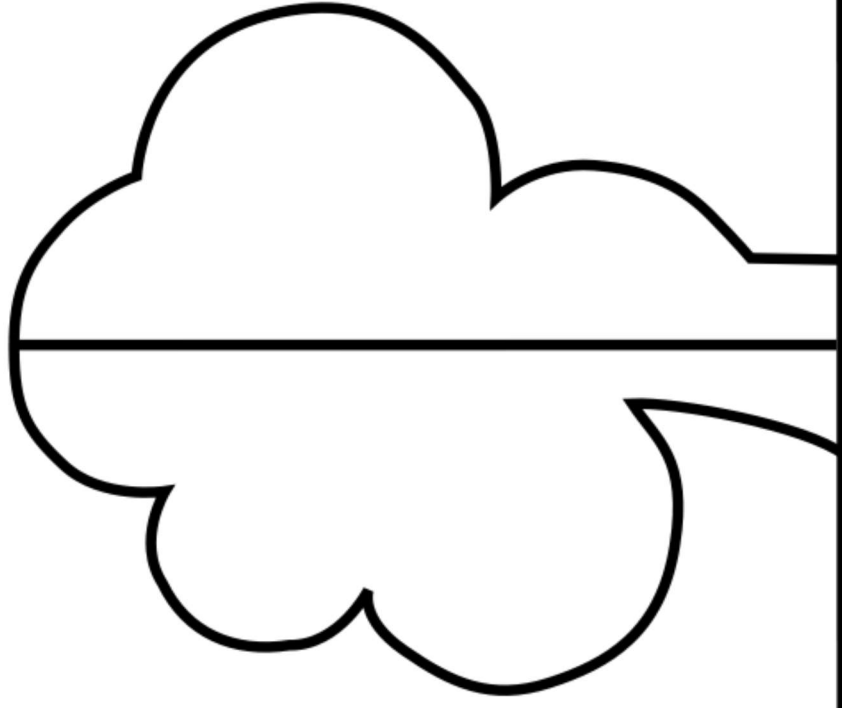


Hydra (invertebrate)

- **Symmetry**, in Biology, the repetition of the parts in an animal or plant in an orderly fashion.
- The vast majority of animals, however, exhibit a definite symmetrical form. Four such patterns of symmetry occur among animals: spherical, radial, biradial, and bilateral.



SYMMETRIC



ASYMMETRIC

Asymmetry

- Asymmetry is an important and widespread trait, having evolved numerous times in many organisms and at many levels of organisation.
- Asymmetry means absence of symmetry.
- Some animals are not divisible into equivalent halves, called asymmetrical animals. Their symmetry type is asymmetry.

- Example: Many protozoans
- Most snails, because of torsion during development, show remarkable asymmetry in the shell and in the internal organs.
- Fiddler crabs have one big claw and one small claw.



Asymmetry

cannot be divided into mirror images

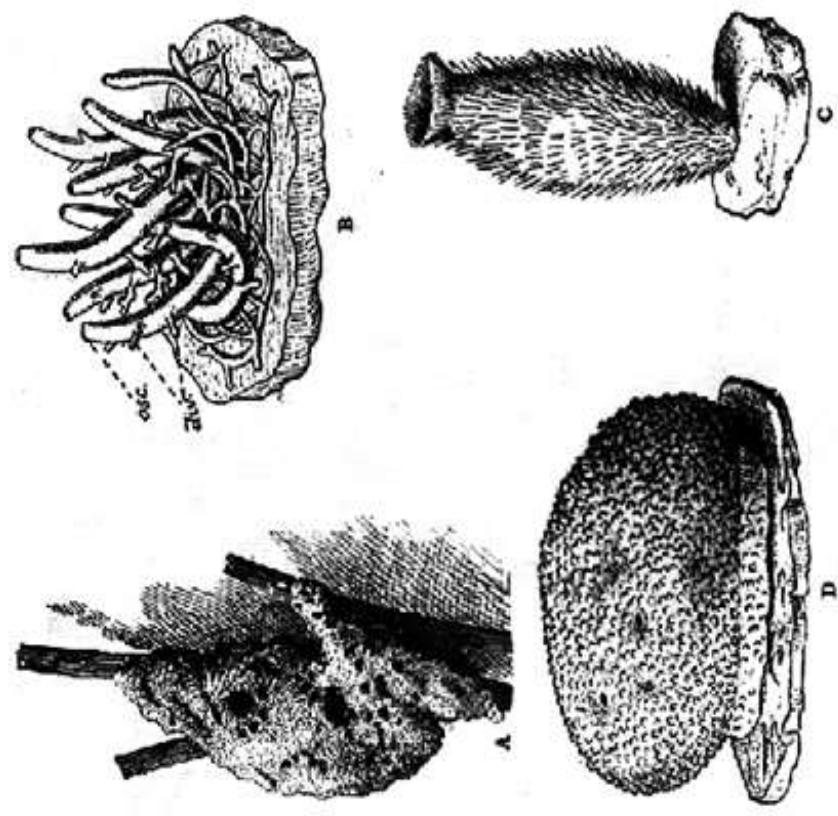
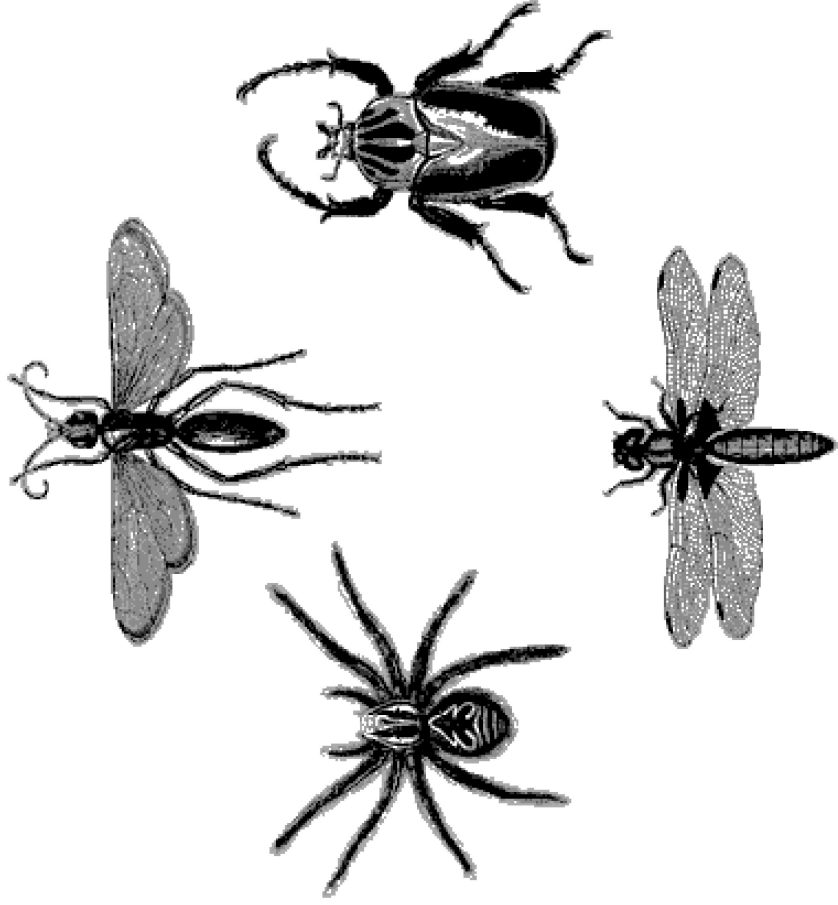


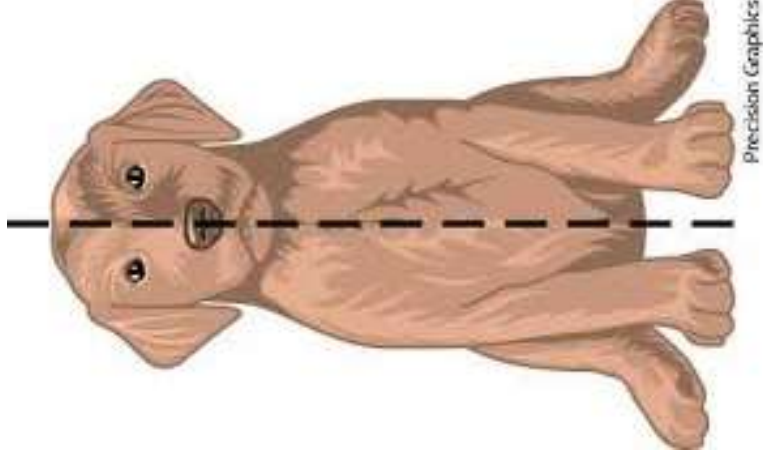
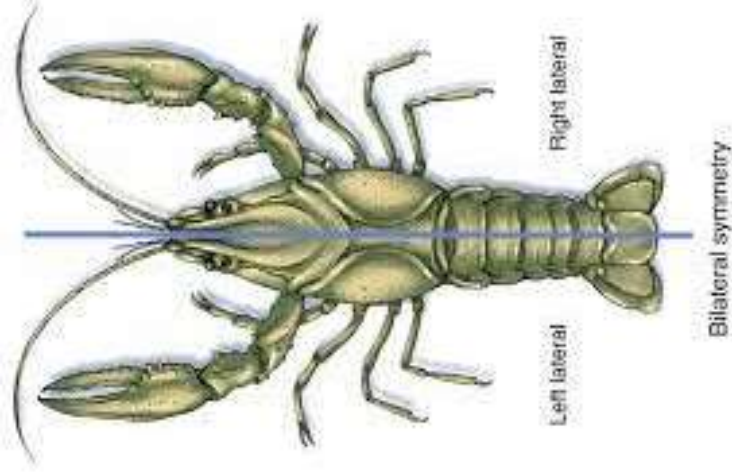
FIG. 127. — Types of sponges.
A, fresh-water sponge; B, a simple-colonial marine sponge; C, a simple solitary marine sponge; D, a bath sponge. (After various authors.)

Bilateral Symmetry:

A single plane divides body into two mirror images



- Bilateral symmetry is characteristic of the vast majority of animals, including insects, fishes, amphibians, reptiles, birds, mammals, and most crustaceans.



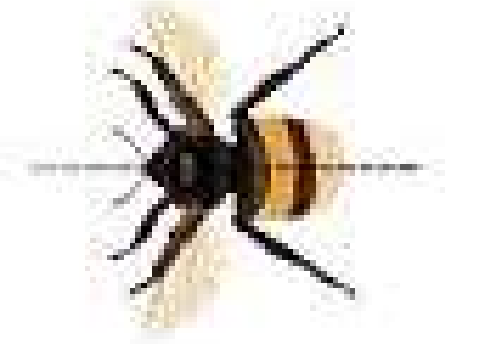
Insects exhibit bilateral symmetry



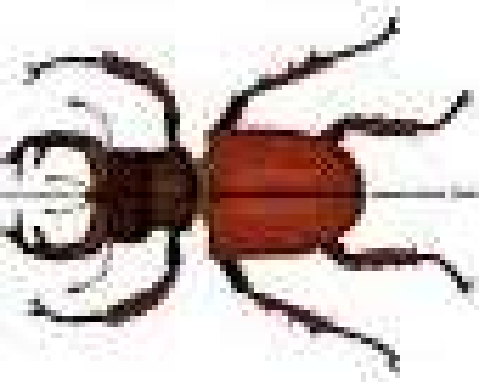
Butterfly



Ladybird



Bee



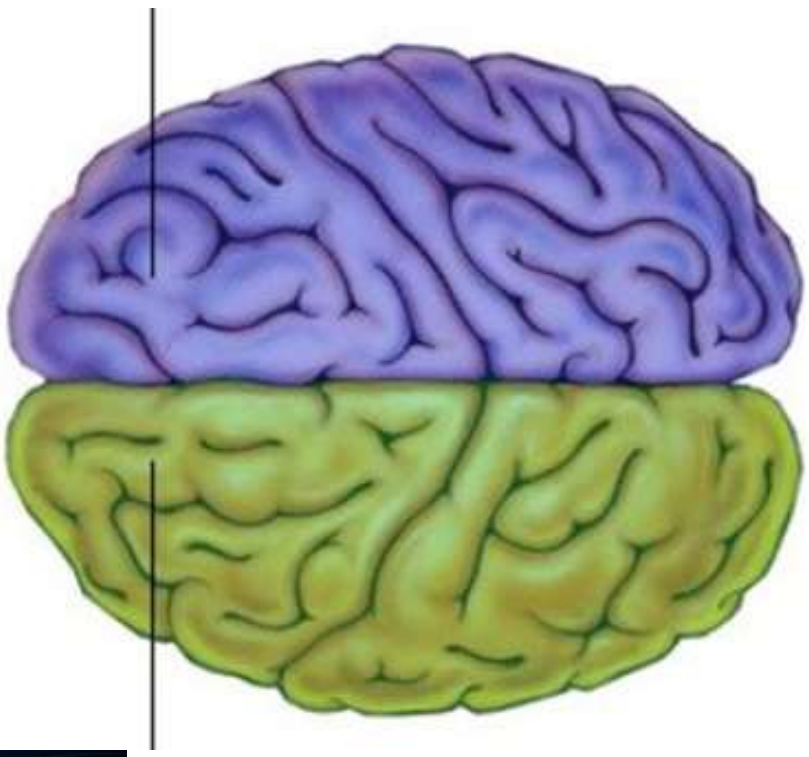
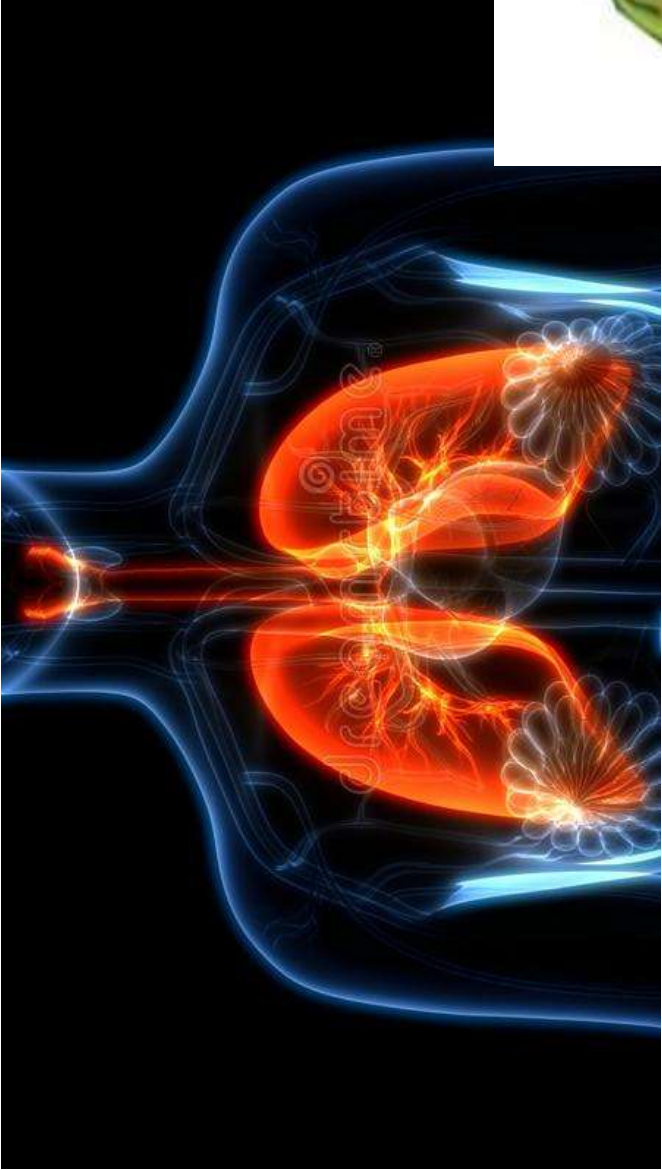
Beetle

Reflection Symmetry

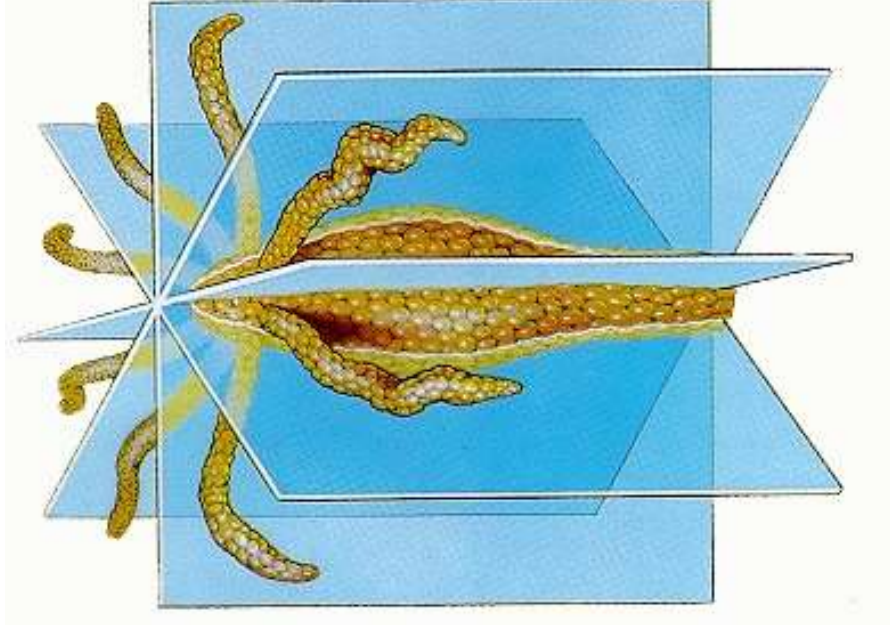
Identify the symmetry —



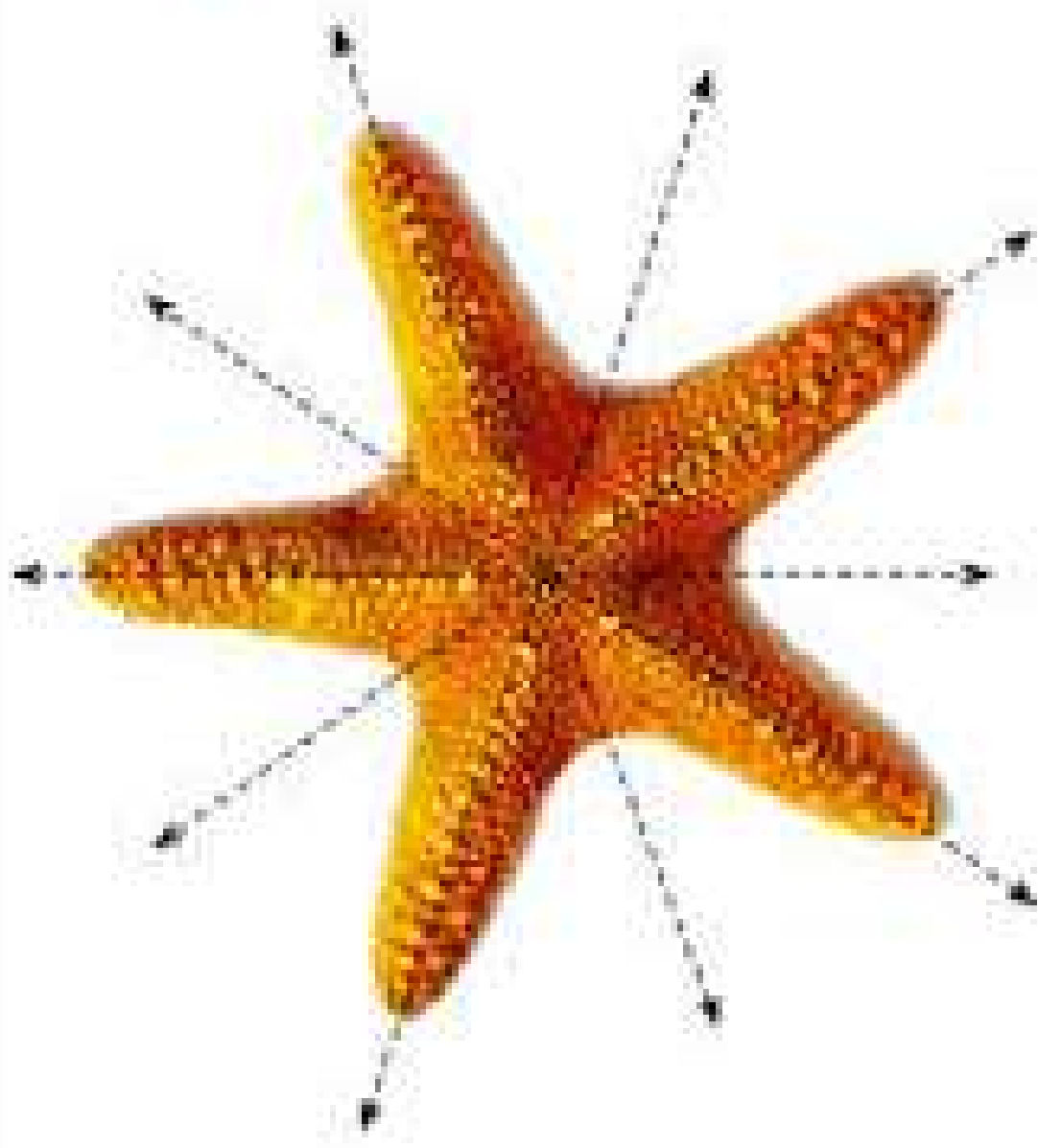
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Radial Symmetry:
can be cut in half along many planes
that allow for nearly identical halves



- A basic body plan in which the organism can be divided into similar halves by passing a plane at any angle along a central axis,
- This is the characteristic of sessile and bottom-dwelling animals, as the sea anemone and starfish.



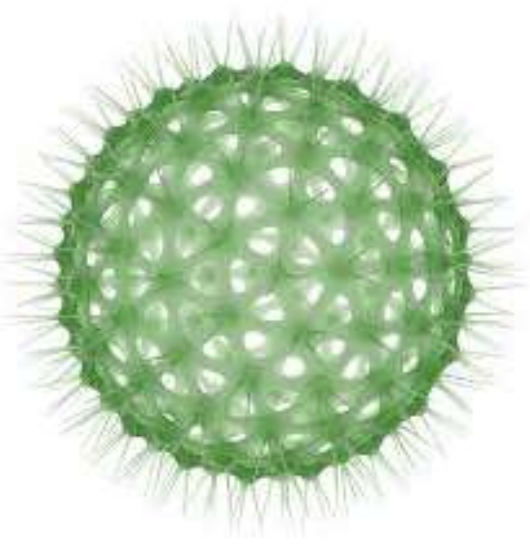
Biradial symmetry

- Biradial symmetry is found in organisms which show morphological features (internal or external) of both bilateral and radial symmetry.
- Ctenophoras
- Polyp of Hydra



Spherical symmetry

- Spherical symmetry is characterized by the ability to draw an endless, or great but finite, number of symmetry through the center of body.
- True spherical symmetry is not found in animal body plans.
- Organisms which show approximate spherical symmetry include the freshwater green algae *Volvox*.



What type of symmetry do each of the following exhibit?



1.



2.



3.



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4.



5.



6.



7.



8.



9.



*Thank
You!*

